

A PROJECT REPORT
ON
Titanic Survival Prediction using
Logistic Regression

**G.L. Bajaj Institute of Technology and
Management Greater Noida**



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1. Introduction

The **Titanic Survival Prediction** project is a machine learning classification task that aims to predict whether a passenger survived the Titanic disaster based on demographic and travel-related information. The Titanic dataset is one of the most widely used datasets in data science and machine learning because it provides a practical example for understanding **binary classification**, where the target variable has two possible outcomes: **Survived (1)** and **Did Not Survive (0)**.

Predicting passenger survival involves analyzing various factors such as age, gender, passenger class, family size, fare paid, and embarkation port. By identifying the relationship between these features and survival, machine learning models can estimate the probability of survival for new passengers. Such predictive models help demonstrate how historical data can be transformed into meaningful insights for decision-making.

In this project, **Logistic Regression** is used as the primary classification algorithm. Unlike Logistic Regression, which predicts continuous values, Logistic Regression estimates the probability of an event occurring and classifies observations into one of two categories. It is widely used because it is simple, efficient, interpretable, and performs well on binary classification problems.

The project follows a complete machine learning workflow that includes dataset understanding, exploratory data analysis (EDA), data preprocessing, model building, evaluation, and business insight generation. The trained model is assessed using standard classification metrics such as **Accuracy, Precision, Recall, F1 Score**, and the **Confusion Matrix** to evaluate its predictive performance.

Overall, this project demonstrates how machine learning techniques can be applied to solve real-world classification problems. In addition to building a predictive model, it provides insights into the factors that influenced passenger survival and highlights how data-driven analysis can support better planning and decision-making in transportation and safety management.

2. Problem Statement

The **Titanic disaster** remains one of the most well-known maritime tragedies in history, resulting in significant loss of life. Passenger survival was influenced by several factors, including age, gender, passenger class, fare, family size, and embarkation port. Analyzing these factors can help identify the characteristics that contributed to survival and improve the understanding of passenger safety.

The objective of this project is to develop a **Logistic Regression** model that predicts whether a passenger survived the Titanic disaster using demographic and travel-related information. Since the target variable has two possible outcomes—**Survived (1)** and **Did Not Survive (0)**—the problem is formulated as a **binary classification** task.

The model is trained using historical Titanic passenger data and evaluated using classification metrics such as **Accuracy, Precision, Recall, F1 Score**, and the **Confusion Matrix**. The findings of this project can provide valuable insights into the factors that most strongly influenced passenger survival and support data-driven decision-making for improving safety planning in future maritime transportation. This aligns with the assignment objective of building a binary classification model using Logistic Regression for the Titanic dataset.

The main objectives of this project are:

- To analyze the Medical Cost Personal Dataset and understand its characteristics.
- To perform Exploratory Data Analysis (EDA) for identifying relationships between variables.
- To preprocess the data using encoding, feature scaling, and train-test splitting techniques.
- To build and evaluate a Logistic Regression model for predicting titanic survival rate.
- To identify the most influential factors affecting insurance charges and derive meaningful business insights.

3. Dataset Description

The **Titanic Dataset** is a well-known benchmark dataset used for binary classification problems in machine learning. It contains information about passengers who traveled on the RMS Titanic, including demographic details, travel information, and whether they survived the disaster. The dataset is publicly available on **Kaggle** and is widely used for learning and evaluating classification algorithms. The objective of this project is to predict passenger survival using these features.

The target variable in the dataset is **Survived**, where:

- **0** – Passenger did not survive
- **1** – Passenger survived

The dataset contains both **numerical** and **categorical** features.

Table 3.1: Description of Dataset Features

Feature	Description	Data Type
PassengerId	Unique identification number assigned to each passenger	Numerical
Survived	Target variable indicating passenger survival (0 = No, 1 = Yes)	Categorical (Binary)
Pclass	Passenger ticket class (1 = First, 2 = Second, 3 = Third)	Categorical
Name	Passenger's full name	Text
Sex	Gender of the passenger	Categorical
Age	Age of the passenger (in years)	Numerical
SibSp	Number of siblings or spouses aboard	Numerical
Parch	Number of parents or children aboard	Numerical
Ticket	Ticket number	Text
Fare	Ticket fare paid by the passenger	Numerical
Cabin	Cabin number assigned to the passenger	Categorical/Text
Embarked	Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)	Categorical

Figure 3.1: First 10 Records of the Titanic Dataset.

```
df.head(10)
```

✓ 0.0s

	Passengerid	Age	Fare	Sex	sibsp	Parch	Pclass	Embarked	Survived
0	1	22.0	7.2500	0	1	0	3	2.0	0
1	2	38.0	71.2833	1	1	0	1	0.0	1
2	3	26.0	7.9250	1	0	0	3	2.0	1
3	4	35.0	53.1000	1	1	0	1	2.0	1
4	5	35.0	8.0500	0	0	0	3	2.0	0
5	6	28.0	8.4583	0	0	0	3	1.0	0
6	7	54.0	51.8625	0	0	0	1	2.0	0
7	8	2.0	21.0750	0	3	1	3	2.0	0
8	9	27.0	11.1333	1	0	2	3	2.0	1
9	10	14.0	30.0708	1	1	0	2	0.0	1

Figure 3.2: Structure and Data Types of the Dataset.

```
df.info
```

✓ 0.0s

```
<bound method DataFrame.info of
0      1  22.0  7.2500  0      1  0      0      0
1      2  38.0 71.2833  1      1  0      0      0
2      3  26.0  7.9250  1      0  0      0      0
3      4  35.0 53.1000  1      1  0      0      0
4      5  35.0  8.0500  0      0  0      0      0
...
1304    1305 28.0  8.0500  0      0  0      0      0
1305    1306 39.0 108.9000  1      0  0      0      0
1306    1307 38.5  7.2500  0      0  0      0      0
1307    1308 28.0  8.0500  0      0  0      0      0
1308    1309 28.0 22.3583  0      1  0      0      0

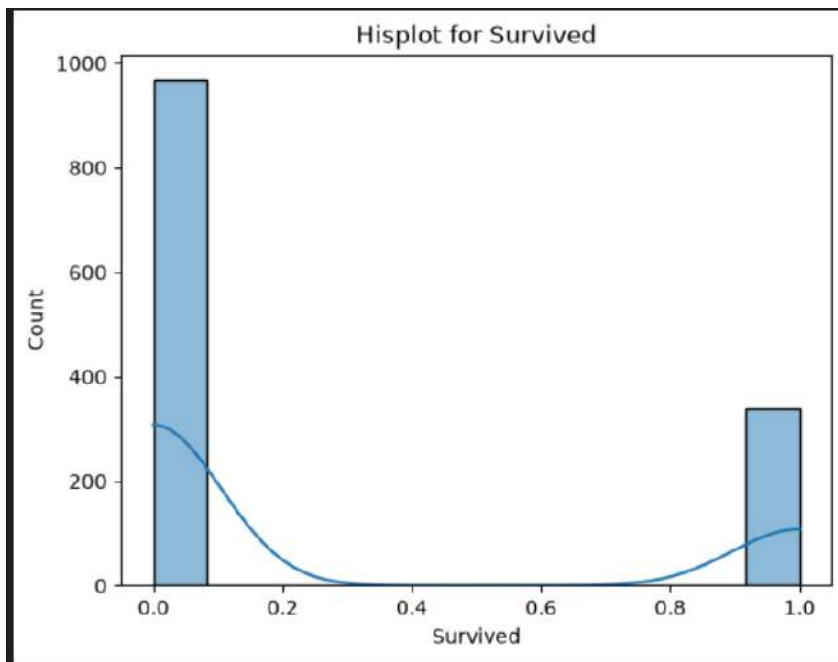
zero.4  ... zero.12 zero.13 zero.14 pclass zero.15 zero.16 \
0      0  ...      0      0      0      3      0      0
1      0  ...      0      0      0      1      0      0
2      0  ...      0      0      0      3      0      0
3      0  ...      0      0      0      1      0      0
4      0  ...      0      0      0      3      0      0
...
1304    0  ...      0      0      0      3      0      0
1305    0  ...      0      0      0      1      0      0
1306    0  ...      0      0      0      3      0      0
1307    0  ...      0      0      0      3      0      0
1308    0  ...      0      0      0      3      0      0
...
1306    2.0      0      0      0
1307    2.0      0      0
1308    0.0      0      0

[1309 rows x 28 columns]>
```

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset, identify relationships among variables, detect outliers, and analyze the distribution of features before building the prediction model. Various visualization techniques were used to gain insights into the data and determine which factors significantly influence medical insurance charges.

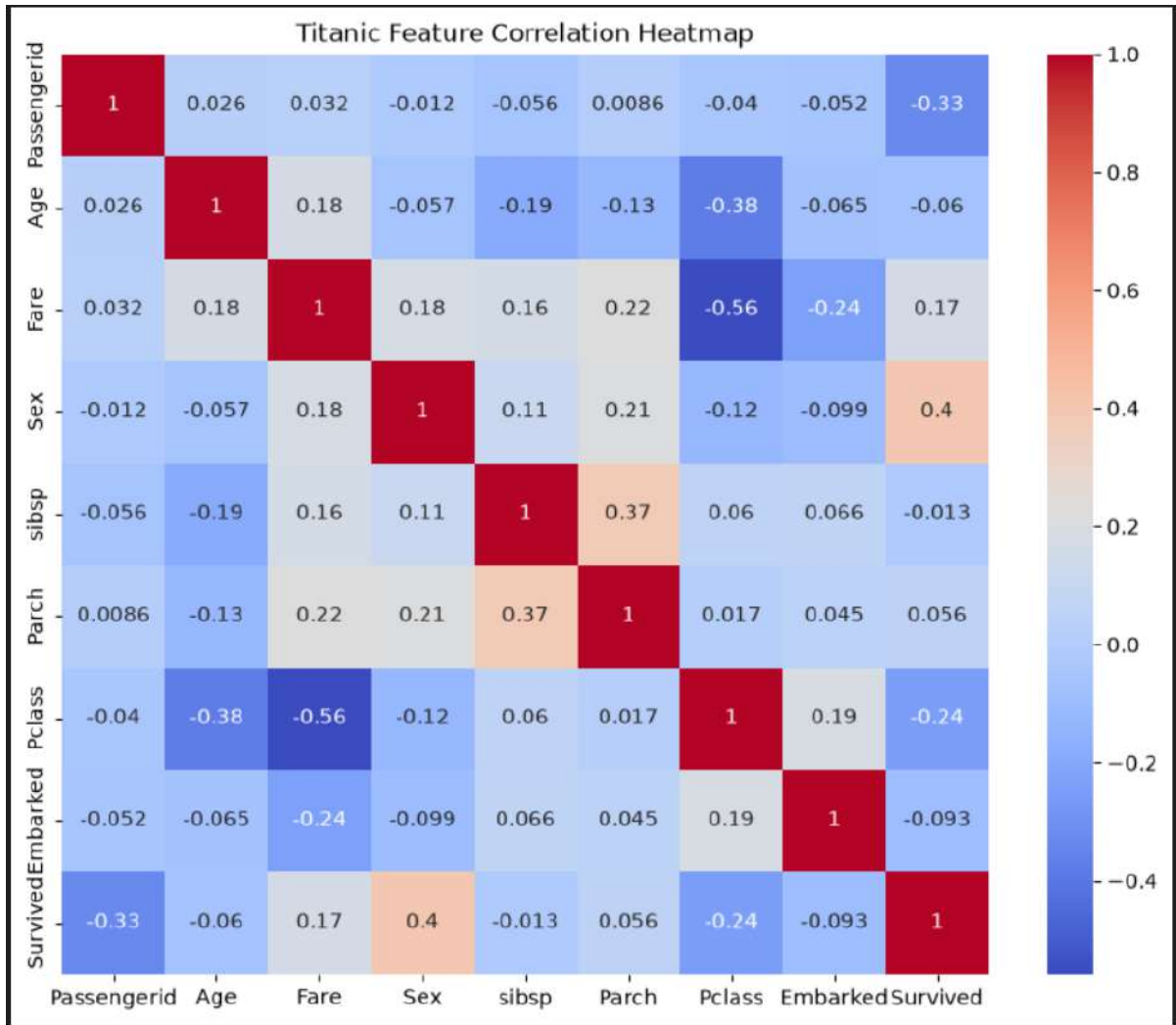
4.1 Class Distribution



Observation:

The dataset is slightly imbalanced, with more passengers not surviving than surviving. Although the imbalance is not severe, it should be considered while evaluating the classification model.

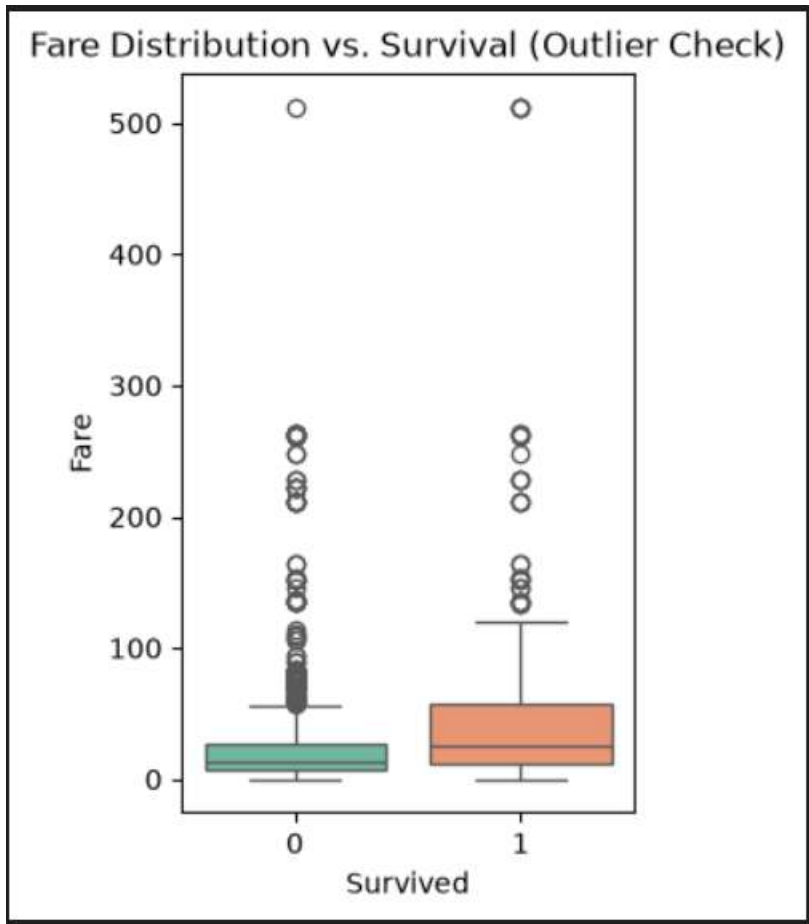
4.2 Correlation Analysis



Observation:

The heatmap indicates that features such as **Fare**, **Passenger Class (Pclass)**, and **Age** show noticeable relationships with the target variable. These features are expected to contribute significantly to predicting passenger survival.

4.3 Outlier Analysis

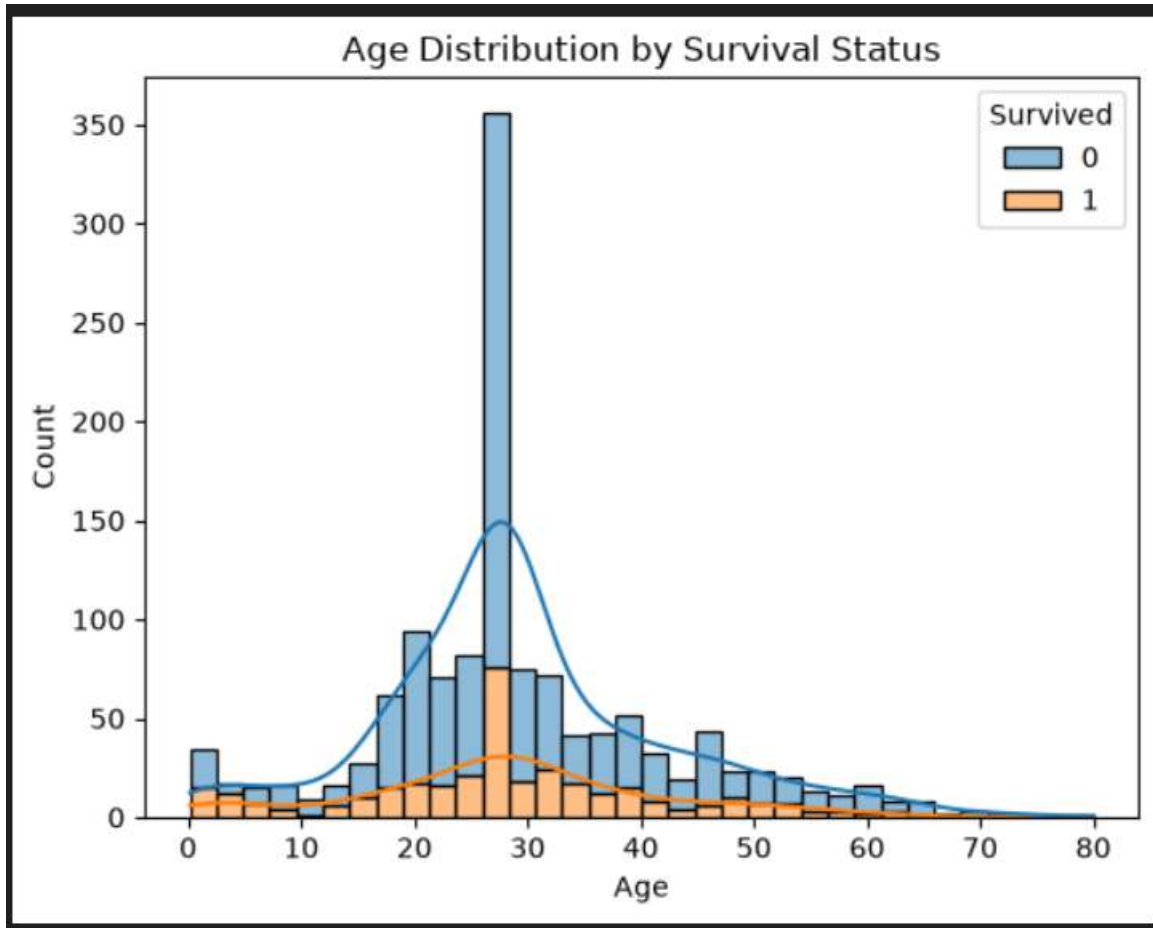


Observation:

The above boxplot helps us understand how fare influenced their survival.

- The Boxes: The colored boxes represent where the vast majority of passengers fall. We can see that the orange box (Survived) sits slightly higher than the green box (Died), meaning survivors generally paid higher fares.
- The Dots (Outliers): The black circles stretching all the way up to \$500 are extreme outliers. These were incredibly wealthy passengers who paid massive premiums for their tickets.

4.4 Relationship Between Age and Survival rate



Observation:

The analysis shows that **younger passengers had a relatively higher survival rate** compared to older passengers. Children were more likely to survive due to strict enforcement of the "women and children first" maritime protocol, while survival generally decreased with increasing age. This suggests that age was an important factor influencing passenger survival and should be considered a valuable predictor in the classification model.

4.5 Summary of EDA

The Exploratory Data Analysis provided valuable insights into the Titanic dataset and the factors affecting passenger survival. The dataset was found to be **slightly imbalanced**, with more passengers not surviving than surviving. Analysis showed that **Passenger Class, Age, and Fare** had a noticeable influence on survival. Younger passengers generally exhibited a higher survival rate, while the **Fare** feature contained several outliers due to a few passengers paying exceptionally high ticket prices. These observations highlighted the importance of data preprocessing, including handling missing values, encoding categorical variables, and feature scaling, before training the Logistic Regression model.

5. Data Preprocessing

Data preprocessing is an important step in the machine learning pipeline that prepares the dataset for model training. It improves data quality and ensures that the model learns meaningful patterns from the data. The following preprocessing steps were performed in this project:

5.1 Missing Value Handling

The dataset was checked for missing values using the `isnull()` function. Only missing values were found in column `Embarked`; therefore, two imputation or data removal was required.

5.2 Feature Encoding

The categorical features (**Sex**, **Survived**, **Pclass**) were converted into numerical form using encoding techniques. Since machine learning algorithms cannot process categorical data directly, encoding was necessary to make these features suitable for model training.

5.3 Feature Scaling

Numerical features were standardized using the **StandardScaler**. Feature scaling ensures that all numerical variables have a similar scale, preventing features with larger values from dominating the learning process. This also improves the performance and stability of regression models.

5.4 Train-Test Split

The dataset was divided into **80% training data** and **20% testing data** using the `train_test_split()` function. The training dataset was used to train the model, while the testing dataset was used to evaluate its performance on unseen data.

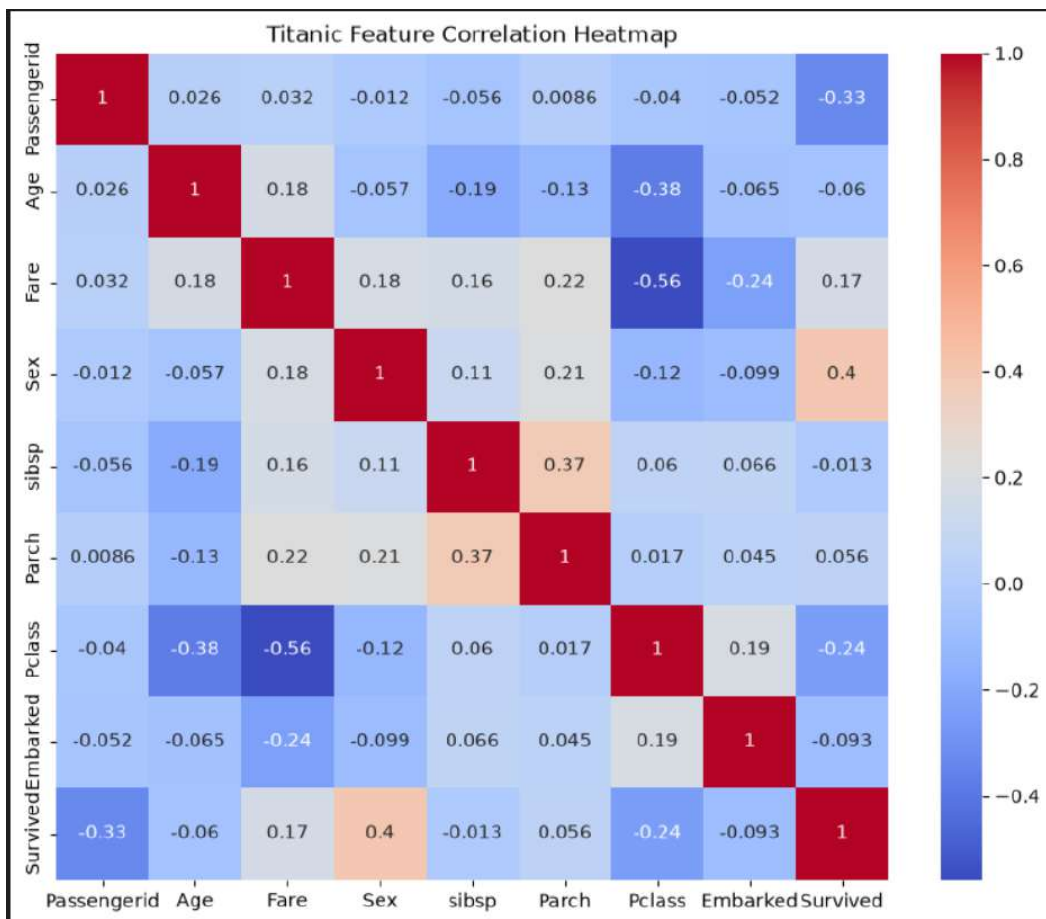
6. Feature Engineering

Feature engineering is the process of transforming raw data into meaningful features that improve the performance of machine learning models. In this project, feature engineering primarily involved preparing the existing features for model training rather than creating new variables.

The categorical features, namely **Sex**, **Survived**, and **Pclass**, were converted into numerical representations using encoding techniques so that they could be processed by the Logistic Regression model. Numerical features were standardized using **StandardScaler**, ensuring that all features were on a similar scale and preventing variables with larger values from disproportionately influencing the model.

No additional features were created because the existing dataset contained sufficient information for predicting medical insurance charges. The transformed features were then used to train and evaluate the regression models.

Figure 6.1: Dataset after Feature Engineering and Transformation



7. Model Building

After preprocessing the dataset, a **Logical Regression** model was developed using the **Scikit-learn** library to predict medical insurance charges. Logistic Regression was selected because the target variable (**survived**) is continuous, making it a suitable algorithm for regression problems. The model aims to establish a Logistic relationship between the independent variables (Passengerid, Age, Fare, Sex, sibsp, Parch, Pclass, Embarked) and the target variable (survived).

The preprocessed dataset was divided into training and testing sets using an 80:20 ratio. The model was trained on the training dataset using the `fit()` method and then used to predict insurance charges for the testing dataset using the `predict()` method. The model learned the contribution of each feature by calculating coefficients and an intercept, which represent the influence of each independent variable on the predicted insurance charges.

The trained model was then evaluated using standard regression metrics to measure its prediction accuracy and overall performance.

Table 7.1: Logical Regression Model Coefficients.

	<i>feature</i>	<i>coefficient</i>
0	<i>Passengerid</i>	<i>-1.190506</i>
3	<i>Sex</i>	<i>1.070612</i>
6	<i>Pclass</i>	<i>-0.795319</i>
1	<i>Age</i>	<i>-0.403090</i>
4	<i>sibsp</i>	<i>-0.288025</i>
7	<i>Embarked</i>	<i>-0.082728</i>
2	<i>Fare</i>	<i>0.030080</i>
5	<i>Parch</i>	<i>-0.003781</i>

8. Model Evaluation

The performance of the Logistic Regression model was evaluated using standard regression metrics to determine how accurately it predicts medical insurance charges. These metrics measure the difference between the actual insurance charges and the values predicted by the model.

Table 8.1: Performance Metrics of Logistic Regression

Metric	Value
Accuracy	83.20610687022901
F1 Score	0.6206896551724138
Recall	0.5142857142857142
Precision	0.7835929767120722

Table 8.2: Confusion Matrix

182	10
34	36

Table 8.3: Classification Report

	Precision	recall	F1-score	support
0	0.84	0.95	0.89	192
1	0.78	0.51	0.62	70
accuracy			0.83	262
macro avg	0.81	0.73	0.76	262
weighted avg	0.83	0.83	0.82	262

8.1 Performance Analysis

- **Accuracy (83.21%)** indicates that the model correctly classified approximately 83% of all passengers, demonstrating good overall performance.
- **Precision (78.26%)** shows that when the model predicts a passenger survived, it is correct about 78% of the time. This indicates that the model produces relatively few false positive predictions.
- **Recall (51.43%)** indicates that the model correctly identified only about 51% of the actual survivors. This suggests that a significant number of survivors were misclassified as non-survivors.
- **F1 Score (62.07%)** represents the balance between precision and recall. The moderate F1 score indicates that while the model makes reliable positive predictions, it misses a considerable number of actual survivors.

Overall, the evaluation results indicate that the Logistic Regression model provides reliable predictions and captures the relationship between the input features and medical insurance charges effectively.

10. Business Insights

The Logistic Regression model provides valuable insights into the factors that influenced passenger survival during the Titanic disaster. These findings can help maritime safety organizations understand passenger risk factors and develop strategies to improve safety measures in future emergencies.

Key Business Insights

1. Passenger Class Significantly Influenced Survival

Passengers traveling in **First Class** had a higher probability of survival than those in **Second** and **Third Class**. This suggests that access to emergency resources and cabin location may have played an important role during evacuation.

2. Gender Had a Major Impact

The analysis indicates that **female passengers were more likely to survive than male passengers**. This supports the historical evacuation policy of prioritizing women during rescue operations.

3. Age Influenced Survival Probability

Children and younger passengers generally had a higher chance of survival than older passengers. This suggests that age was an important factor in determining rescue priority and survival outcomes.

4. Fare and Socioeconomic Status

Passengers who paid **higher ticket fares** generally showed better survival rates. Since higher fares were often associated with first-class travel, this indicates that socioeconomic factors indirectly affected survival.

5. Important Predictors

The Logistic Regression model identified **Passenger Class (Pclass)**, **Sex**, **Age**, and **Fare** as some of the most influential features for predicting passenger survival. These variables contributed significantly to the model's classification decisions.

Recommendations for Maritime Safety

Based on the analysis, the following recommendations can help improve passenger safety in future maritime travel:

- Ensure **equal access to lifeboats and emergency exits** for passengers across all travel classes.

- Implement **clear evacuation procedures** that prioritize vulnerable passengers, including children, older adults, and individuals requiring assistance.
- Conduct **regular emergency drills** for both passengers and crew to improve preparedness during emergencies.
- Improve **emergency communication systems** to provide timely and accurate evacuation instructions.
- Use **data-driven risk assessment models** to support emergency planning, passenger management, and resource allocation.

Overall Business Impact

The developed Logistic Regression model successfully predicts passenger survival with **an overall accuracy of approximately 83%**, demonstrating its ability to identify survival patterns from historical passenger data. The insights obtained from this model can assist maritime authorities in improving evacuation planning, optimizing safety policies, and making evidence-based decisions to enhance passenger safety during future maritime incidents.

11. Conclusion

This project successfully developed a **Titanic Survival Prediction** system using the **Logistic Regression** algorithm. The dataset was analyzed through exploratory data analysis (EDA), followed by data preprocessing techniques such as handling missing values, feature selection, encoding categorical variables, feature scaling, and train-test splitting. These steps prepared the data for building an effective classification model.

The Logistic Regression model achieved an **accuracy of 83.21%**, demonstrating good overall performance in predicting passenger survival. The model also achieved a **precision of 78.26%**, **recall of 51.43%**, and an **F1 Score of 62.07%**, indicating that it was effective at identifying non-survivors while leaving room for improvement in detecting all surviving passengers. Overall, the evaluation results show that Logistic Regression is a reliable and interpretable algorithm for binary classification problems such as Titanic survival prediction.

The project also identified key factors influencing passenger survival, including **passenger class, gender, age, and fare**. These insights can help maritime safety organizations better understand historical survival patterns and support the development of improved emergency response strategies.

11.1 Future Scope

The performance of the model can be further improved by:

- Performing **hyperparameter tuning** to optimize the Logistic Regression model.
- Applying advanced classification algorithms such as **Random Forest**, **XGBoost**, or **Support Vector Machine (SVM)**.
- Creating additional features through feature engineering to enhance predictive performance.
- Using techniques to address class imbalance, such as **SMOTE** or class weighting.
- Deploying the trained model as a web application to provide real-time survival probability predictions.